LET US DREAM
Software Requirements Specification (SRS)

Prepared for: Let Us Dream / DREAMS mentoring and development programme
Purpose: define the complete product scope, modules, workflows, data entities, roles, reports, AI-enabled transcript generation, repository features, and non-functional requirements for implementation

Product Type	Web and mobile-ready mentoring, assessment, reflective reporting, volunteer documentation, and progress tracking platform
Primary Users	Admin, Mentors, Community Leaders, Faculty Leaders, Strategic Team, Principals, Volunteers, Mentees
Core Outcome	Track three-year growth journey, structured session reflections, attendance, action tracking, approvals, volunteer contribution history, and dashboards
Long-term Goal	Provide an integrated digital infrastructure for DIP, YCLP, LUD, volunteer development, reporting, and institutional decision support across all locations

1. Introduction

The Let Us Dream application is a structured mentoring and developmental tracking platform intended to support mentors, mentees, volunteers, programme coordinators, school leaders, and approval authorities across a multi-year intervention journey. The system is expected to replace fragmented manual tracking with a central, role-based platform for planning sessions, documenting observations, monitoring attendance, tracking action items, reviewing growth over time, preserving reflective reports, and generating evidence-backed volunteer development outputs.

The existing workflow indicates a process starting from login, mentor access, chapter selection by school, year-wise mentee mapping, assessment planning, and review/approval by Community Leaders and Faculty Leaders. The tracker workbook shows that the programme measures different developmental domains for Year 1, Year 2, and Year 3 using structured ratings, mentor remarks, and action plans. The reflexive reporting form adds a reporting layer for learning, evidence uploads, endorsements, and feedback. The long-term digital vision further requires explicit coverage of DIP-YCLP reflective reporting across all locations, a mentee tracking system, an AI-assisted volunteer transcript generator, public profiles, work diaries, a shared data repository, and notification services. This SRS consolidates all of these into one implementation-ready baseline.

1.1 Product Vision

Build a single digital platform that is simple enough to be completed in 5 to 10 minutes per mentee per session, mostly structured through checkboxes, sliders, and guided fields, while still enabling intentional reflection, risk spotting, action planning, longitudinal tracking, volunteer documentation, and executive visibility across all programme locations.

1.2 Business Objectives

- Standardise DIP, YCLP, and LUD reporting across years, chapters, schools, and locations.
- Reduce manual Excel-, paper-, email-, and form-based documentation effort.
- Provide reliable progress visibility to mentors, leadership, and programme coordinators.
- Enable approvals, evidence-based reviews, and future-stage readiness decisions.

- Create structured records for audit, programme evaluation, volunteer development, and reporting.
- Generate personalized volunteer transcripts as a structured alternative to traditional recommendation letters.
- Build a scalable long-term digital infrastructure for Let Us Dream.

1.3 In Scope

- Role-based login and permissions.
- Mentor, mentee, volunteer, leadership, and admin modules.
- School/chapter/year/location mapping.
- Session planning, attendance, assessment, and observation capture.
- DIP-YCLP reflective reporting across all programme locations.
- Volunteer work diary and contribution logging.
- AI-assisted volunteer transcript generation.
- Public and restricted profile management.
- Shared repository for documents, files, evidence, and resources.
- Approval workflow.
- Dashboards and downloadable reports.
- Bulk upload of mentee master data via Excel/CSV.
- Three-year progression history and term snapshots.
- Notification and reminder services.

1.4 Out of Scope for Phase 1

- Unrestricted peer-to-peer social networking features.
- Advanced autonomous AI scoring decisions without human review.
- Offline-first operation in no-network environments.
- Financial accounting and fee management.
- External public publishing of sensitive mentee information without explicit governance approval.

2. Stakeholders and User Roles

Role	Main Responsibilities in System
System Administrator	Configure schools, chapters, academic years, locations, users, templates, rating scales, topics, transcript templates, repository categories, approval rules, reports, imports, and master data.
Mentor	View allocated mentees, conduct sessions, mark attendance, record structured observations, assign actions, submit reflective reports, maintain diaries where applicable, upload materials, and monitor progress.
Volunteer	Maintain profile, work diary, reflective submissions, repository uploads where permitted, and access generated transcript history.
Mentee	View profile, assigned chapter/year, session history, materials, action commitments, attendance summary, and approved feedback appropriate to access level.
Community Leader	Review session details, progress records, reflective reports, and approvals for their chapter, school, or location.
Faculty Leader	Review and approve assessment details, transcript recommendations where configured, flag issues, and monitor chapter performance.
Strategic Team / Principal	Read-only executive visibility across schools, years, locations, trends, risks, volunteer engagement, and programme outputs.
Data Entry / Coordinator	Bulk upload mentees, assign mentor mapping, correct records, manage operational data quality, and assist with repository/resource upkeep.

3. High-Level Workflow

Step	System Behaviour
1. Login	Users authenticate and land on role-specific dashboards.
2. Mentor and volunteer space	Mentors and volunteers access profiles, attendance, reflective reports, transcripts, work diary, topic materials, repository assets, and notifications.
3. Location and chapter selection	Programme is organised by location, school, chapter, and year with configurable structure.
4. Year allocation	Mentees are mapped to Year 1, Year 2, or Year 3.
5. Mentee import	School mentee names and details are uploaded via spreadsheet and validated before publishing.
6. Session setup	Assessment topic and session date are created based on year, location, programme, and session topic.
7. Session recording	Mentors capture attendance, ratings, reflections, barriers, emotional readiness, empowerment, and action tracker outcomes.
8. Reflective and diary capture	DIP/YCLP/LUD reflective reports and volunteer work diaries are submitted with evidence and metadata.
9. Review and approval	Community Leaders and Faculty Leaders review and approve or return records.
10. Transcript generation	Approved mentor and volunteer data are aggregated for AI-assisted transcript drafting and export.
11. Dashboarding	Overall dashboards show location-level, chapter-level, mentee-level, volunteer-level, yearly, and cumulative progress.
12. Continuity	The system preserves continuity across the full three-year journey and volunteer engagement history.

4. Product Modules

Module	Description
Authentication & Access Control	Secure login, password reset, session management, role-based permissions, and optional SSO integration.
User & Master Data Management	Manage schools, chapters, locations, years, topics, roles, mentors, volunteers, mentees, academic cycles, and approval routing.
Mentor Workspace	Daily operational area for session planning, mentee mapping, attendance, assessment entry, action tracking, and reflections.
Mentee Progress Tracking	Store year-wise competency ratings, mood changes, remarks, action plans, and readiness indicators.
Session Assessment Engine	Dynamic form builder that serves year-specific rubrics, required fields, structured reflections, and validation rules.
DIP-YCLP Reflective Reporting	Capture structured reflective reports for DIP, YCLP, and LUD activities from all locations with evidence and endorsement.
Volunteer Transcript Generator	Compile approved evidence from mentoring, reporting, diaries, attendance, leadership input, and AI-assisted summaries into personalized volunteer transcripts.
Volunteer Public Profiles	Manage public and internal volunteer-facing profiles, contribution summaries, badges, milestones, and visibility controls.
Work Diary System	Allow volunteers and authorised mentors to maintain day-wise or week-wise work diaries linked to sessions, initiatives, events, and outputs.
Shared Data Repository	Central repository for materials, evidence, forms, guidelines, training resources, outputs, and approved artefacts with search and permissions.
Approval Workflow	Multi-level review by Community Leaders and Faculty Leaders with status changes and audit trails.
Reports & Dashboards	Operational, analytical, and executive-level views with export to PDF/Excel/CSV.
Notifications &	System alerts, reminders, pending approval notifications, overdue action

Communication	reminders, transcript status notifications, and announcement broadcast.
Import/Export Utilities	Bulk upload mentees and mapping data from Excel and export filtered reports for management use.
API & Integration Layer	Expose APIs for web, mobile, and future institutional integrations.

5. Detailed Functional Requirements

ID	Requirement
FR-01	The system shall support secure login for Admin, Mentor, Volunteer, Mentee, Community Leader, Faculty Leader, Strategic Team, and Principal roles.
FR-02	The system shall support password reset, account activation, and account deactivation.
FR-03	The system shall allow admins to create and manage schools, chapters, locations, academic years, programmes, and topics of assessment.
FR-04	The system shall allow bulk upload of mentee master data through Excel/CSV with validation preview and error reporting.
FR-05	The system shall allow mapping of mentees to school, chapter, location, year, mentor, and programme cycle.
FR-06	The system shall support mentor and volunteer profile management and assignment workflows.
FR-07	The system shall provide a role-specific dashboard immediately after login.
FR-08	The system shall allow configuration of up to 15 chapters or more without code change.
FR-09	The system shall allow session creation based on year, topic, programme, location, and scheduled date.
FR-10	The system shall allow attendance marking per session and per mentee.
FR-11	The system shall support year-specific assessment rubrics for Year 1, Year 2, and Year 3.
FR-12	The system shall capture beginning mood and end mood for each mentee session record.
FR-13	The system shall support rating fields using a configurable Likert scale, default 1 to 5.
FR-14	The system shall compute average scores automatically per mentee per session and per term.
FR-15	The system shall store mentor remarks and action plan for each recorded session.
FR-16	The system shall capture structured observation fields including dream or interest expressed, external barriers, emotional readiness, and empowerment level.
FR-17	The system shall support an action tracker capturing whether action was agreed last session and whether it was completed, partially completed, or not attempted.
FR-18	The system shall capture narrative reflection fields including positive growth noticed, concern or risk to monitor, and recommended follow-up action.
FR-19	The system shall support periodic DREAMS Progress Snapshot recording top growth areas, top challenges, recommended focus, and readiness for next stage.
FR-20	The system shall allow mentors to upload topic materials and supporting resources.
FR-21	The system shall allow authorized users to upload photos, outputs, files, and links as evidence.
FR-22	The system shall support reflective reports with programme tag, location, email, name, register number or volunteer ID, class or role, date, duration, activity, endorser, learnings or reflections, attachments, and suggestions.

FR-23	The system shall route completed records to Community Leaders and Faculty Leaders for review and approval.
FR-24	The system shall allow reviewers to approve, reject, request changes, or comment on submitted records.
FR-25	The system shall preserve a complete audit trail of who created, edited, reviewed, and approved each record.
FR-26	The system shall provide chapter-level dashboards for mentors and leaders.
FR-27	The system shall provide mentee-level cumulative progress across all three years.
FR-28	The system shall provide filters by school, chapter, location, mentor, volunteer, year, term, session type, programme, and approval status.
FR-29	The system shall export reports in Excel and PDF formats.
FR-30	The system shall support notifications for pending assessments, pending approvals, missing attendance, overdue follow-up actions, and transcript publication.
FR-31	The system shall support broadcast notifications or messages to chapter members where authorized.
FR-32	The system shall provide search by mentee name, register number, school, chapter, volunteer, mentor, or location.
FR-33	The system shall allow controlled edit windows and record locking after approval.
FR-34	The system shall support archival of past academic years while keeping records readable.
FR-35	The system shall generate consolidated reports for Principals and Strategic Team users.
FR-36	The system shall expose APIs for mobile client or future integrations.
FR-37	The system shall support DIP-YCLP reflective reporting from all programme locations under one common reporting model.
FR-38	The system shall allow configurable report templates for DIP, YCLP, and LUD while retaining a common approval and analytics structure.
FR-39	The system shall maintain volunteer work diaries with date, activity, location, linked session or initiative, duration, narrative entry, evidence, and review status.
FR-40	The system shall provide a personalized volunteer transcript generator that uses approved reporting data, attendance, work diary data, mentoring records, recognitions, and AI-assisted summarization.
FR-41	The system shall allow authorized users to review, edit, approve, and publish the AI-generated transcript before final release.
FR-42	The system shall support multiple transcript templates for different purposes such as volunteer service summary, leadership transcript, and programme completion transcript.
FR-43	The system shall support volunteer public profiles with configurable visibility, biography, contribution summary, achievements, approved photos, and programme associations.
FR-44	The system shall provide internal-only profile views for leadership containing more detailed contribution and performance history than public views.
FR-45	The system shall provide a shared data repository with role-based folders, file categories, tagging, metadata, search, and access rules.
FR-46	The system shall allow repository assets to be linked to sessions, reports, diaries, profiles, and transcript evidence.
FR-47	The system shall support location-wise analytics and data segregation while also enabling central network-wide reporting.
FR-48	The system shall support AI-assisted summary suggestions for reflective reports and transcripts, while requiring human review before approval.

6. Year-wise Assessment Rubrics

Year	Indicative dimensions to capture
Year 1	Session Engagement; Self-Awareness and Growth; Emotional and Behavioral Regulation; Initiative and Responsibility; Mentorship and Reflection.
Year 2	Receiving Feedback; Help-Seeking Behavior; Team Work; Respect; Accountability.
Year 3	Leadership Management; Decision Making; Time Management; Value-Based Action; Conflict Navigation.

6.1 Common Session Fields

Field Group	Fields
Identity	Mentee Name, Mentor Name, Grade, School, Chapter, Location, Year, Register No., Class.
Session Metadata	Session or camp type, programme tag, date, topic or theme, duration, location or mode, created by, approval status.
Affective Inputs	Beginning mood, end mood, emotional readiness, empowerment level.
Observation Inputs	Dream or interest reinforced, external barriers, positive growth noticed, concern or risk, follow-up action.
Developmental Scoring	Year-specific rubric dimensions and weighted or unweighted average.
Action Tracker	Action agreed last session, attempted or completed status, action owner, due date, remarks.
Evidence	Photos, files, links, reflective report uploads, diary artefacts, endorser remarks.

7. Additional Functional Coverage Required by Long-term Digital Vision

7.1 DIP - YCLP Reflective Reporting

The platform shall explicitly support structured reflective reporting for DIP, YCLP, and LUD activities across all locations. Each report shall be tagged by programme, location, chapter, reporting period, facilitator or mentor, and linked activity. The system shall permit both structured fields and narrative reflections so that reporting remains analyzable without losing qualitative learning.

- Location-wise report submission and central consolidation.
- Configurable templates per programme with common data model.
- Evidence uploads, endorsement fields, and review workflow.
- Searchable archive of approved reports for institutional learning and audit.

7.2 Mentee Tracking System

The existing mentee tracking baseline is retained and strengthened. The platform shall act as a post-session logging tool that enables mentors to record observations and progress of each mentee and consolidate this data across the full network.

- Per-session observations, ratings, emotional indicators, and action tracking.
- Cross-session history, term snapshots, and year-on-year comparison.
- Location-wise and network-wide analytics for mentors and leadership.
- Risk and follow-up visibility through reminders and dashboards.

7.3 Volunteer Transcript Generator

The platform shall support a volunteer transcript generator as an alternative to traditional letters of recommendation. This module shall compile structured contribution evidence from DIP-YCLP reporting, attendance, diaries, approved activities, mentor inputs, recognitions, and AI-assisted summary drafting.

- Transcript drafting shall remain editable by authorized reviewers before release.
- Final output shall support PDF generation, digital verification metadata where required, and archival storage.
- Transcript templates shall be configurable for different volunteer pathways or programme roles.
- The transcript module shall not issue fully autonomous judgments without human approval.

7.4 Additional Features

- Public profiles for volunteers with controlled visibility and approved highlights.
- Work diaries for continuous contribution logging and reflective documentation.
- Shared data repository for documents, evidence, and resources.
- Notification system for reminders, approvals, transcript status, and engagement prompts.

8. Representative User Stories

As a…	I want to…	So that…
Mentor	open a pre-filled session form for my mapped mentees	I can complete assessment quickly during or right after the session.
Coordinator	upload a spreadsheet of school mentees	I can reduce manual data entry and maintain standard formatting.
Community Leader	review submitted session records and reflective reports for my scope	I can ensure programme quality and compliance.
Faculty Leader	approve or return records and transcript drafts with comments	I can maintain academic and pastoral oversight.
Principal	view consolidated dashboards by school, location, and year	I can understand outcomes and intervention needs.
Volunteer	maintain my diary and access my approved contribution transcript	I can document and present my work credibly.
Strategic Team Member	compare DIP and YCLP reporting patterns across locations	I can identify programme strengths, gaps, and support needs.

9. Workflow Specifications

9.1 Mentee Onboarding Workflow

Admin or coordinator uploads mentee records. The system validates required columns, duplicates, mentor assignment, school or chapter mapping, location mapping, and year allocation. Approved uploads create or update mentee master records and assignment maps.

9.2 Session Planning Workflow

Mentor selects chapter, location, year, topic, and date or uses a pre-scheduled session created by admin. The system loads the correct assessment template, mapped mentees, and pending actions from the previous session.

9.3 Session Capture Workflow

Mentor marks attendance, enters structured ratings, completes observations and reflection, attaches evidence where needed, and saves as draft or submits for review.

9.4 Review and Approval Workflow

After submission, the record moves to Community Leader review. The reviewer may approve, return for revision, or escalate. Depending on configuration, Faculty Leader performs final approval. Approved records become read-only to standard users.

9.5 Longitudinal Tracking Workflow

At term end or camp end, the mentor or coordinator records a DREAMS Progress Snapshot. The platform compares current and prior periods, rolls the history into a cumulative timeline, and carries the mentee forward to the next year when promoted.

9.6 Reflective Reporting Workflow

Users with permission submit activity-level reflective reports with programme tag, learnings, evidence, endorser, and suggestions. These reports remain linked to the relevant session, event, or programme period.

9.7 Work Diary Workflow

Volunteer or authorized mentor creates a diary entry linked to a date, location, activity, and optional session or initiative. The entry may be submitted as draft, reviewed if required, and later referenced in transcript generation.

9.8 Transcript Workflow

The system aggregates approved volunteer evidence, prepares an AI-assisted transcript draft, routes it to authorized reviewer(s), and allows final export after approval.

9.9 Repository Workflow

Authorized users upload resources or evidence to the repository with category, tags, and ownership metadata. The system enforces visibility and access controls and allows linking from other modules.

10. Minimum Screens / Pages

Screen	Purpose
Login / Forgot Password	Authentication and recovery.
Dashboard	Role-based overview and alerts.
Mentor Workspace	Assigned mentees, pending sessions, pending actions, notifications.
Volunteer Workspace	Diary, profile, transcript status, approved activities, notifications.
Mentee Directory	Search, filter, view profile, mapping, and history.
Session Entry Form	Rapid structured entry for a session.
Bulk Upload Wizard	Import master data with validation.
Approval Queue	Review submitted records and take actions.
Reflective Report Form	Activity documentation with evidence uploads.
Work Diary Page	Chronological diary entry and history management.
Volunteer Profile Page	Public or restricted profile overview with visibility control.
Transcript Builder / Review Page	AI draft review, manual edits, template selection, and final export.
Repository	Searchable file and resource management.
Reports & Analytics	Operational and executive analytics.
Settings / Master Data	Topics, rubrics, years, chapters, locations, scale definitions, templates.

11. Core Data Entities

Entity	Key Attributes
User	User ID, name, email, mobile, role, status, password or SSO metadata.
Mentor	Mentor ID, linked user, school/chapter/location scope, profile details, active status.
Volunteer	Volunteer ID, linked user, programme role, profile summary, achievements, transcript status, visibility settings.
Mentee	Mentee ID, register no., name, grade, class, school, chapter, year, guardian

	details, active status.
Location	Location ID, name, code, school mapping, active status.
Academic Cycle	Cycle ID, year label, start date, end date, status.
Chapter	Chapter ID, school, location, name, code, leader mapping.
Mentor-Mentee Mapping	Mapping ID, mentor, mentee, effective dates, primary flag.
Topic / Assessment Template	Template ID, year, programme, topic, rubric items, mandatory fields, scoring rules.
Session	Session ID, date, type, topic, chapter, location, year, facilitator, status.
Attendance	Attendance ID, session, mentee, status, timestamp, remarks.
Assessment Record	Record ID, session, mentee, ratings, moods, empowerment, barriers, narrative fields, average score.
Action Item	Action ID, assessment record, action text, owner, due date, status.
Approval Log	Approval ID, record, reviewer, decision, comments, timestamp.
Reflective Report	Report ID, programme, location, activity, duration, endorser, learning text, feedback text.
Diary Entry	Diary ID, volunteer, date, duration, location, linked activity, narrative entry, review status.
Volunteer Transcript	Transcript ID, volunteer, template, generated summary, reviewer notes, approval status, export file reference.
Profile Artifact	Artifact ID, volunteer profile section, image or document reference, visibility status.
Repository Asset	Asset ID, title, category, file path, tags, linked entity, owner, uploader, visibility, version.
Evidence Attachment	Attachment ID, file path, type, size, linked entity, uploader.
Notification	Notification ID, recipient, type, payload, read flag, delivery status.

12. Key Business Rules

Rule ID	Rule
BR-01	A mentee may belong to only one active year at a time within a programme cycle.
BR-02	A mentee may have one primary mentor and optional secondary mentor or coordinator.
BR-03	The displayed assessment rubric shall depend on the mentee year and selected session topic or template.
BR-04	Mandatory fields must be completed before submission for approval.
BR-05	Average score shall be auto-calculated from enabled rubric dimensions.
BR-06	Approved records are locked against standard editing; reopen requires elevated permission.
BR-07	Action items from the previous session must be visible in the next session form.
BR-08	Only authorised reviewers may approve records for scopes within their assigned chapter, school, or location.
BR-09	Uploads containing duplicate register numbers in the same active cycle shall be blocked or merged based on admin rule.
BR-10	All critical actions shall be audit logged.
BR-11	Reflective reports shall always carry programme and location metadata for cross-network reporting.
BR-12	Volunteer transcripts shall be generated only from approved and eligible source records.
BR-13	AI-assisted summaries shall not be published until reviewed and approved by an authorized human reviewer.
BR-14	Public profile content shall display only fields explicitly marked as public.
BR-15	Repository access shall follow role-based and asset-level permissions.

13. Non-Functional Requirements

Category	Requirement
Usability	The session entry experience shall support completion in approximately 5 to 10 minutes per mentee using structured controls and minimal typing.
Accessibility	The UI shall support responsive layouts, keyboard navigation, readable contrast, and clear error messages.
Performance	Common dashboard and form pages should load within 3 seconds under expected load; bulk import validation should provide progress feedback.
Scalability	The platform shall support multiple schools, locations, chapters, cycles, and growing record history without redesign.
Availability	Target operational availability shall be at least 99.5% during programme periods excluding planned maintenance.
Security	Data shall be protected with role-based access control, encrypted transport, secure password handling, and audit trails.
Privacy	Sensitive mentee data shall be visible strictly on a need-to-know basis with configurable field-level visibility where required.
Maintainability	Rubrics, topics, year mappings, approval workflows, profile sections, transcript templates, and repository categories should be configurable through admin screens instead of code changes.
Interoperability	The system should support data import/export and REST APIs for future integration with institutional systems.
Reliability	Autosave drafts and controlled validation shall reduce accidental loss of assessment, report, and diary data.
Backup & Recovery	Daily backups and point-in-time recovery strategy shall be defined before production launch.
Observability	System logs, job logs, import logs, error tracking, transcript generation logs, and approval audit logs shall be available to support operations.

14. Permission Matrix (Baseline)

Capability	Admin	Mentor	Volunteer	Leader/Reviewer
Manage users/master data	Yes	No	No	Limited
View assigned mentees	Yes	Yes	No	Scoped
Create session records	Yes	Yes	No	No
Submit reflective reports	Yes	Yes	Yes	Scoped
Maintain diary	Yes	Optional	Yes	Read/Review
View transcript	Yes	No	Self only	Scoped
Approve/reject records	Configurable	No	No	Yes
Approve transcript draft	Configurable	No	No	Yes
View dashboards	Yes	Yes	Limited self	Yes
Export reports	Yes	Scoped	Limited self	Scoped
Manage public profile visibility	Yes	No	Self with approval rules	Scoped
Upload evidence/materials	Yes	Yes	Limited	Scoped
Configure rubrics/topics/templates	Yes	No	No	No

15. Required Reports and Dashboards

Report / Dashboard	Purpose

Mentor dashboard	Pending sessions, pending approvals, recent mentee activity, overdue actions.
Chapter dashboard	Attendance, score distributions, risk flags, completion rates, approval ageing.
Mentee progress profile	Timeline of scores, moods, action completion, notes, and year-on-year change.
Volunteer dashboard	Diary completion, approved activities, transcript status, contribution summary.
School leadership dashboard	Cross-chapter and cross-location comparison and operational health.
Programme summary report	Counts by year, completion, attendance, average scores, high-risk cases, promoted or completed mentees.
Reflective report register	Event or activity submissions with programme tags, endorsements, and evidence links.
Transcript issuance register	Generated transcript history with status, template, reviewer, and export audit.
Repository usage report	Uploaded assets, downloads, categories, missing metadata, and permission exceptions.
Data quality report	Missing fields, duplicate records, unapproved submissions, and import issues.

16. Integration Requirements

Bulk import from Excel/CSV is mandatory because the workflow explicitly references uploading school mentee names and details through Excel. The application should therefore include a guided upload wizard with downloadable templates, mapping help, validation logs, and rollback for failed batches.

Optional future integrations may include institutional identity provider for SSO, notification gateways for email/SMS/WhatsApp, document signing or verification service for transcripts, and academic systems for student metadata synchronisation.

17. Recommended Technical Architecture

A standard three-tier web architecture is suitable for Phase 1: front-end web client, secure REST API service, and relational database. Mobile app support can be introduced later using the same API layer. File storage should be handled through object storage or secure server file storage with signed-access patterns for attachments and transcript exports.

Recommended baseline stack choices can include React or Vue for front-end, Django / Node.js / Laravel / Spring Boot for backend depending team capability, PostgreSQL or MySQL for relational storage, Redis for queues/cache, and background workers for imports, notifications, and transcript generation jobs.

18. Audit, Compliance, and Data Governance

Every assessment, report, diary, transcript, repository asset, and approval record must preserve created by, created at, updated by, updated at, and approval history. The system should support retention policies by academic cycle. Access to sensitive observations and barriers must be controlled and logged. Public profile publication and transcript sharing shall follow role-based approval and institutional governance policy.

19. Testing and Acceptance

Development should include unit tests, API tests, role permission tests, import validation tests, workflow tests, transcript generation tests, diary and repository tests, and UAT with real mentor and volunteer scenarios. Test data should cover all three years, approval statuses, missing actions, absent mentees, duplicate upload errors, multi-location reporting, and transcript review cases.

19.1 Sample Acceptance Criteria

ID	Acceptance Criterion
AC-01	Admin can upload a valid mentee spreadsheet and see created records after confirmation.
AC-02	Mentor opening a Year 2 session sees the Year 2 rubric, prior action items, and mapped mentees only.
AC-03	A submitted record appears in the correct reviewer queue and cannot be edited by mentor after final approval.
AC-04	Dashboard totals change when new attendance and assessment records are approved.
AC-05	Reflective report form accepts files, captures programme, location, endorser and learnings, and links to the relevant activity.
AC-06	The system can show cumulative progress for a mentee across all three years.
AC-07	A volunteer can maintain a diary entry and submit it for review where configured.
AC-08	Authorized reviewer can generate, edit, approve, and export a volunteer transcript based on approved source data.
AC-09	Repository assets can be uploaded, searched, and accessed only according to assigned permissions.
AC-10	Executive dashboard can show location-wise DIP and YCLP reporting totals without mixing restricted records across permission scopes.

20. Suggested Phase Breakdown

Phase	Focus
Phase 1	Authentication, master data, imports, mentor workspace, year-wise assessments, attendance, approvals, reflective reporting, core dashboards.
Phase 2	Volunteer portal, work diary, repository, transcript generation workflow, enhanced notifications, location analytics.
Phase 3	Mobile app, advanced integrations, predictive risk insights, multilingual support, offline-assisted capture, digital verification enhancements.

21. Conclusion

This specification translates the mentoring workflow, tracker structure, reflective reporting format, and long-term Let Us Dream digital vision into a development-ready baseline. It now explicitly covers the four required strategic components: (1) DIP-YCLP reflective reporting across all locations, (2) mentee tracking as a post-session logging and analytics system, (3) an AI-assisted volunteer transcript generator as an alternative to traditional recommendation letters, and (4) additional modules for public profiles, work diaries, a shared data repository, and notification services. Before development starts, the team should confirm field-level definitions, approval rules, profile visibility rules, transcript templates, and final reporting layouts with programme stakeholders.